The Journal of Open Source Software

UAVTrafficPy: A Python package for analysis and visualization of UAV-based traffic data

Konstantinos Pourgourides [1], **Yiolanda Englezou** [1], **Christos Panayiotou** [1], and **Stelios Timotheou** [1]

1 KIOS Research and Innovation Center of Excellence, and the Department of Electrical and Computer Engineering, University of Cyprus

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 11 October 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

This paper introduces UAVTrafficPy, an open-source Python package for scientific analysis and visualization of UAV-based traffic data, with a focus on urban environments, such as arterial roads and signalized intersections. The package offers tools to extract, reconstruct, and visualize UAV-based traffic information by utilizing only basic inputs such as vehicle IDs, vehicle types, and their time-dependent spatial coordinates. These tools operate across multiple analytical scales: (a) microscopic: focusing on the behaviour and movement of individual vehicles, (b) mesoscopic: examining interactions between successive vehicles, and (c) macroscopic: analyzing traffic dynamics at the intersection level.

Statement of need

The introduction of UAV-based data collection methods in transportation research has unlocked new possibilities which were inaccessible in the past due to the limitations of traditional sensor-based methods, such as loop detectors, static cameras and probe vehicles equiped with tracking technologies (Barmpounakis & Geroliminis, 2020; Krajewski et al., 2018). These conventional methods suffer from different drawbacks such as sparsity of data, limited coverage of the traffic network, high installation and maintenance costs, and the inability to capture and store high quality spatio-temporal information for individual vehicles. Usually, they have to be combined together to compensate for their inadequencies and deliver reliable results (Zhan et al., 2017).

On the other hand, the rapid deployment of coordinated and camera-equipped UAVs above traffic networks enables researchers to overcome these limitations, as they can acquire large volumes of data for all individual vehicles in a potentially large area, with high spatio-temporal resolution (Gupta et al., 2021). This can transform the landscape of traffic analysis in transportation, provided that appropriate tools are developed. UAVTrafficPy aims to bridge the gap between UAV-based data collection and meaningful, hollistic traffic analysis across multiple analytical scales.

State of the Field

Several published works utilize UAV-based traffic data to accomplish specific goals and address particular issues, such as studying vehicle interactions (Li et al., 2025), inferring successive vehicle spacings and their relative speeds (Jiao et al., 2023), calibrating car-following models (Zhang et al., 2022), treating noise and anomalies (Mahajan et al., 2023), investigating lane-related phenomena (Espadaler-Clapés et al., 2023), and studying the resolution of queues at signalized intersections (Hu et al., 2022). Additionally, there are open-source software packages that address video analysis for UAV-based traffic monitoring (Benjdira et al., 2022),

and simulation of traffic data collection via UAVs (Tsioutis et al., 2025). However, to the best of the authors' knowledge, there is currently no work similar to UAVTrafficPy, which offers tools for scientific analysis and visualization of UAV-based traffic data across multiple analytical scales, in a single software package.

Software Functionality

UAVTrafficPy offers tools for extracting, reconstructing and visualizing UAV-based traffic data across multiple analytical scales as follows.

(a) Microscopic: ***focusing on the behavior and movement of individual vehicles.*** Users can reconstruct and visualize trajectories, calculate time-dependent speed and acceleration profiles, compute cumulative distance travelled, visualize space-time diagrams, and detect lane changes.

(b) Mesoscopic: ***examining interactions between successive vehicles.*** Users can compute relative dynamic gaps and speed differences between successive vehicles, two quantities that are commonly used as variables in the calibration of car-following models.

(c) Macroscopic: ***analyzing traffic dynamics at the intersection level.*** Users can calculate turn ratios, extract information on traffic light phases and cycles, and compute queue-wise information such as the number of queued vehicles, the queue length, and finally the queue dissipation time.

UAVTrafficPy was developed and tested in Python 3.12; it is a standalone package, and not an extension of any already existing software. It requires only minimal information to operate, such as vehicle IDs, vehicle types, and 2D spatial coordinates in the WGS84 system, labeled by the time stamp for every vehicle. Once a user provides this information in the appropriate format, which is thoroughly explained in the walkthrough on the repository's frontpage, UAVTrafficPy can perform a wide range of tasks. Examples include the reconstruction of vehicle trajectories based on routes (Figure 1), the visualization of traffic light phases (Figure 2) of a signalized intersection, and the extraction of the lateral distribution of vehicles in a road (Figure 3) for the recognition of lane characteristics. Figures 1-3 where generated using data from the file `20181024_d2_0900_0930.csv` of the pNEUMA open-source dataset (Barmpounakis & Geroliminis, 2020).

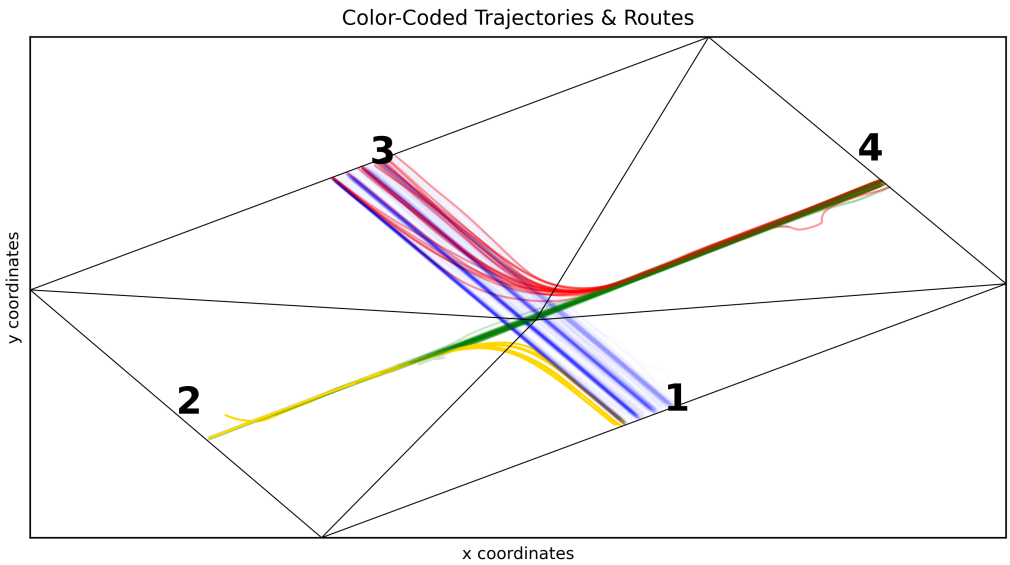

Figure 1: Reconstruction of UAV-based vehicle trajectories based on their routes within a signalized intersection.

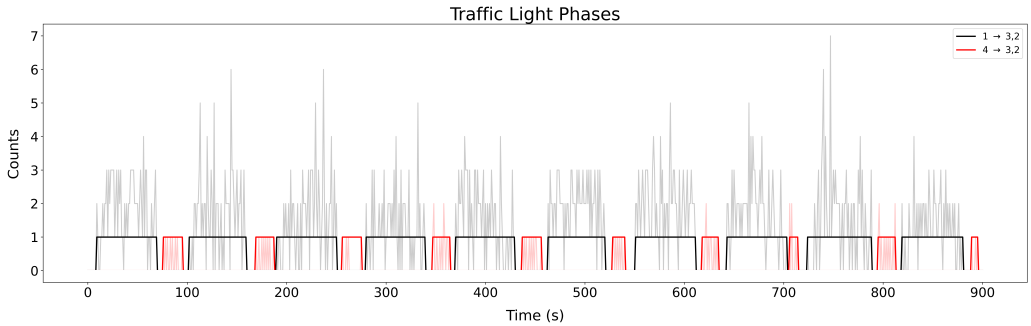

Figure 2: Visualization of the traffic light phases for a signalized intersection.

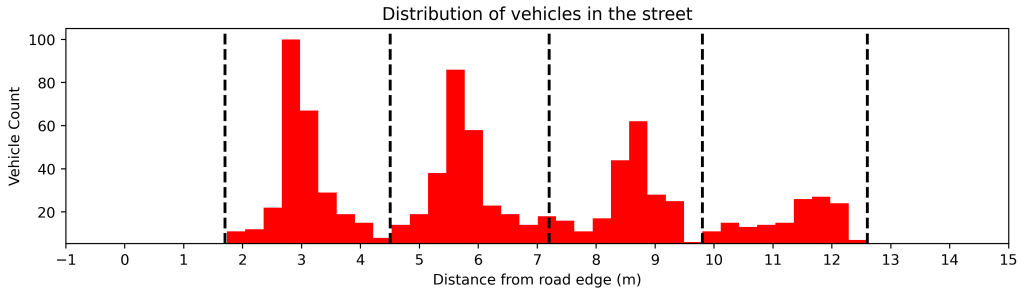

Figure 3: Extraction of the lateral distribution of vehicles in a road.

Acknowledgements

This work is supported by the European Union (i. ERC, URANUS, No. 101088124 and, ii. Horizon 2020 Teaming, KIOS CoE, No. 739551), and the Government of the Republic of Cyprus through the Deputy Ministry of Research, Innovation, and Digital Strategy. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

References

Barmpounakis, E., & Geroliminis, N. (2020). On the new era of urban traffic monitoring with massive drone data: The pNEUMA large-scale field experiment. *Transportation Research Part C: Emerging Technologies*, *111*, 50–71. https://doi.org/https://doi.org/10.1016/j.trc.2019.11.023

Benjdira, B., Koubaa, A., Azar, A. T., Khan, Z., Ammar, A., & Boulila, W. (2022). TAU: A framework for video-based traffic analytics leveraging artificial intelligence and unmanned aerial systems. *Engineering Applications of Artificial Intelligence*, *114*, 105095. https://doi.org/https://doi.org/10.1016/j.engappai.2022.105095

Espadaler-Clapés, J., Barmpounakis, E., & Geroliminis, N. (2023). Empirical investigation of lane usage, lane changing and lane choice phenomena in a multimodal urban arterial. *Transportation Research Part A: Policy and Practice*, *172*, 103674. https://doi.org/https://doi.org/10.1016/j.tra.2023.103674

Gupta, A., Afrin, T., Scully, E., & Yodo, N. (2021). Advances of UAVs toward future transportation: The state-of-the-art, challenges, and opportunities. *Future Transportation*, *1*(2), 326–350. https://doi.org/10.3390/futuretransp1020019

Hu, S., Zhou, Q., Li, J., Wang, Y., Roncoli, C., Zhang, L., & Lehe, L. (2022). High time-resolution queue profile estimation at signalized intersections based on extended kalman filtering. *IEEE Transactions on Intelligent Transportation Systems*, *23*(11), 21274–21290. https://doi.org/10.1109/TITS.2022.3173515

Jiao, Y., Calvert, S. C., van Cranenburgh, S., & van Lint, H. (2023). Inferring vehicle spacing in urban traffic from trajectory data. *Transportation Research Part C: Emerging Technologies*, *155*, 104289. https://doi.org/https://doi.org/10.1016/j.trc.2023.104289

Krajewski, R., Bock, J., Kloeker, L., & Eckstein, L. (2018). The highD dataset: A drone dataset of naturalistic vehicle trajectories on german highways for validation of highly automated driving systems. *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, 2118–2125. https://doi.org/10.1109/ITSC.2018.8569552

Li, Y., Jiang, Y., & Wu, X. (2025). TrajPT: A trajectory data-based pre-trained transformer model for learning multi-vehicle interactions. *Transportation Research Part C: Emerging Technologies*, *171*, 105013. https://doi.org/https://doi.org/10.1016/j.trc.2025.105013

Mahajan, V., Barmpounakis, E., Alam, Md. R., Geroliminis, N., & Antoniou, C. (2023). Treating noise and anomalies in vehicle trajectories from an experiment with a swarm of drones. *IEEE Transactions on Intelligent Transportation Systems*, *24*(9), 9055–9067. https://doi.org/10.1109/TITS.2023.3268712

Tsioutis, C., Makridis, C., & Timotheou, S. (2025). SUAVPy: A SUMO plugin for UAV-based ground traffic sensing. *SUMO Conference Proceedings*, *6*, 65–77. https://doi.org/10.52825/scp.v6i.2610

Zhan, X., Zheng, Y., Yi, X., & Ukkusuri, S. V. (2017). Citywide traffic volume estimation using trajectory data. *IEEE Transactions on Knowledge and Data Engineering*, *29*(2), 272–285. https://doi.org/10.1109/TKDE.2016.2621104

Zhang, Y., Chen, X., Wang, J., Zheng, Z., & Wu, K. (2022). A generative car-following model conditioned on driving styles. *Transportation Research Part C: Emerging Technologies*, *145*, 103926. https://doi.org/https://doi.org/10.1016/j.trc.2022.103926

DRAFT